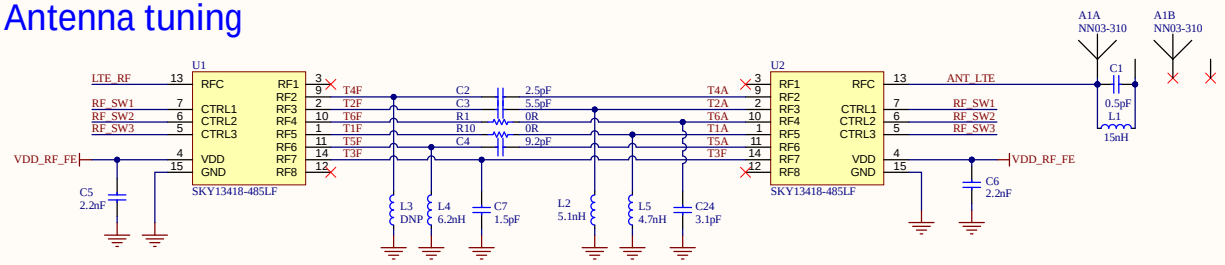
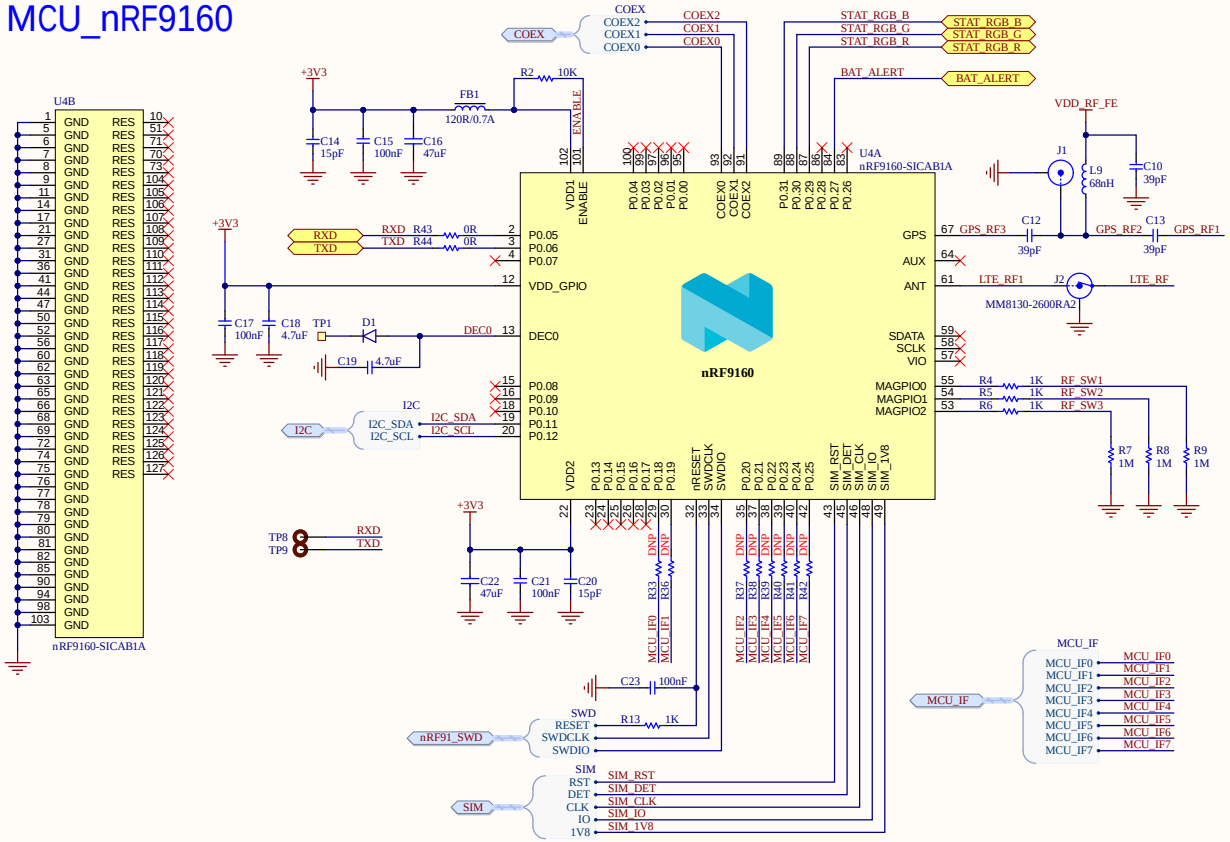
Antenna tuning

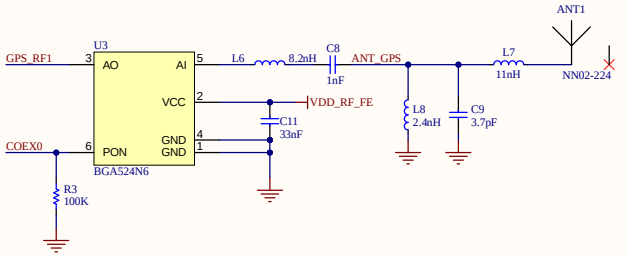


RF_SW3	RF_SW2	RF_SW1	State	Band	Frequency
0	0	0	RF1 - RFC	Not used	Not used
0	0	1	RF5 - RFC	13U/D, 28D	746 MHz - 803 MHz
0	1	0	RF3 - RFC	12U/D, 17U/D, 28U	698 MHz - 748 MHz
				1U/D, 2U/D, 3U/D, 4U/D, 25U/D	1710 MHz - 2200 MHz
0	1	1	RF7 - RFC	5D, 20U, 26D	824 MHz - 894 MHz
1	0	0	RF2 - RFC	8U/D	880 MHz - 960 MHz
1	0	1	RF6 - RFC	5U, 20D, 26U	791 MHz - 849 MHz
1	1	0	RF4 - RFC	Not used	Not used
1	1	1	RF8 - RFC	Not used	Not used

MCU_nRF9160



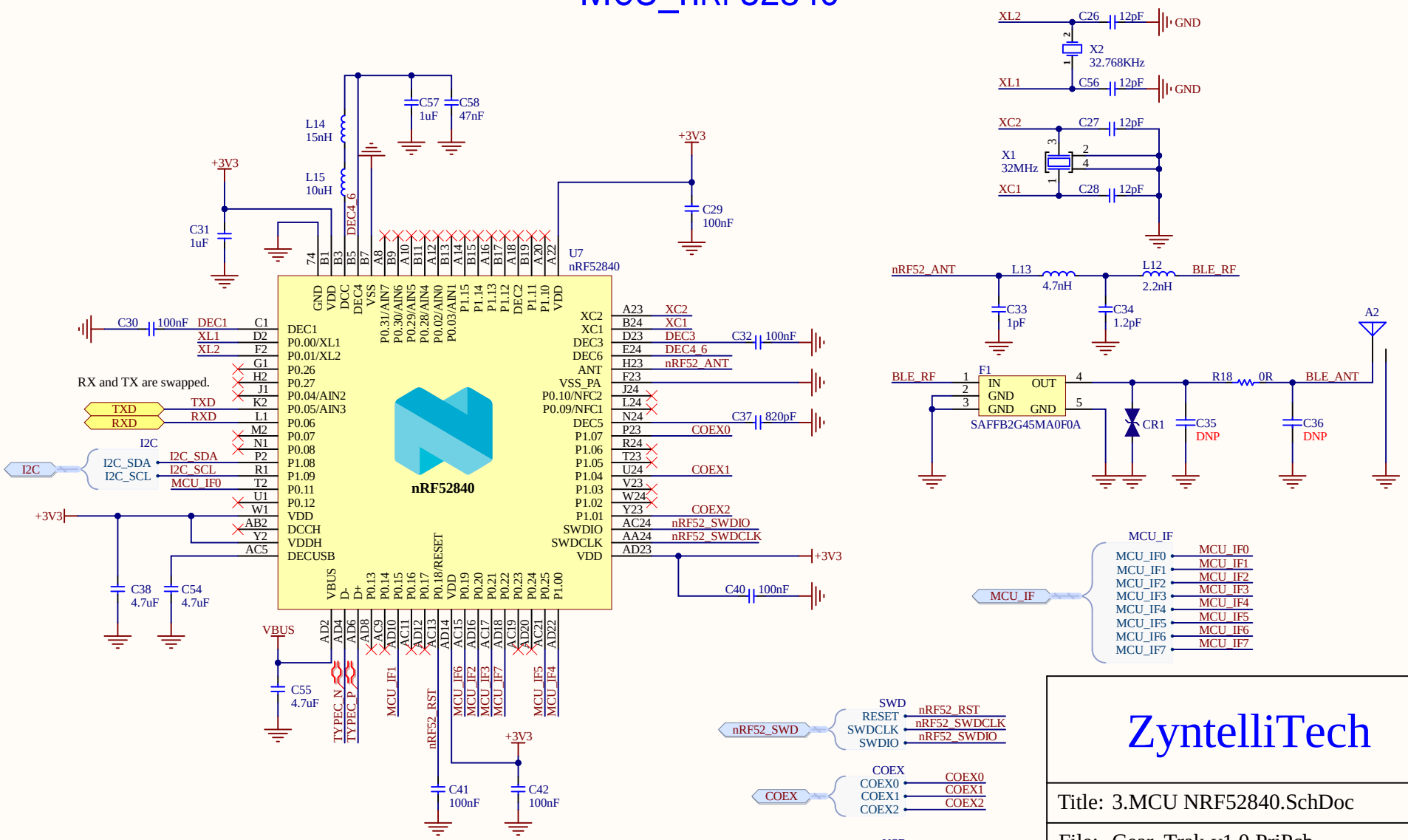
GPS LNA



ZyntelliTech

Title: 4.Microcontroller-NRF9160.SchDoc	
File: Gear_Trak-v1.0.PrjPcb	
Date: 7/16/2025	Page Num: 2
Drawn By: Huzaifa Ali	Page Total: *

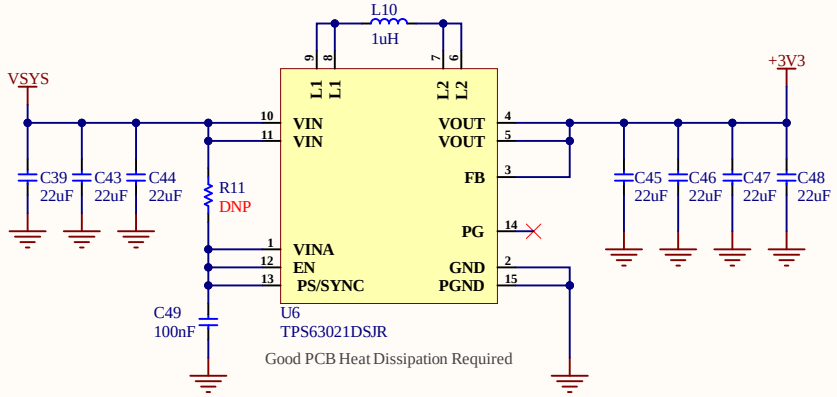
MCU_nRF52840



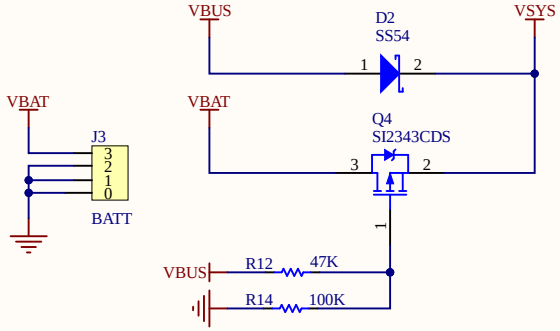
ZyntelliTech

Title: 3.MCU NRF52840.SchDoc	
File: Gear_Trak-v1.0.PrjPcb	
Date: 7/16/2025	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 6

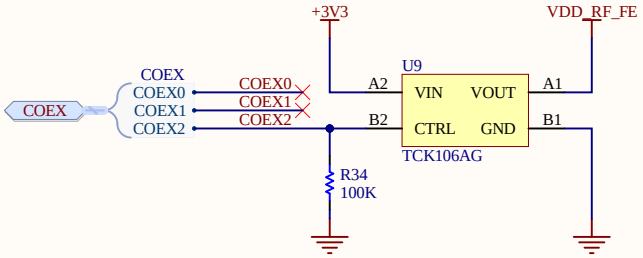
3.3V BUCK BOOST CONVERTER



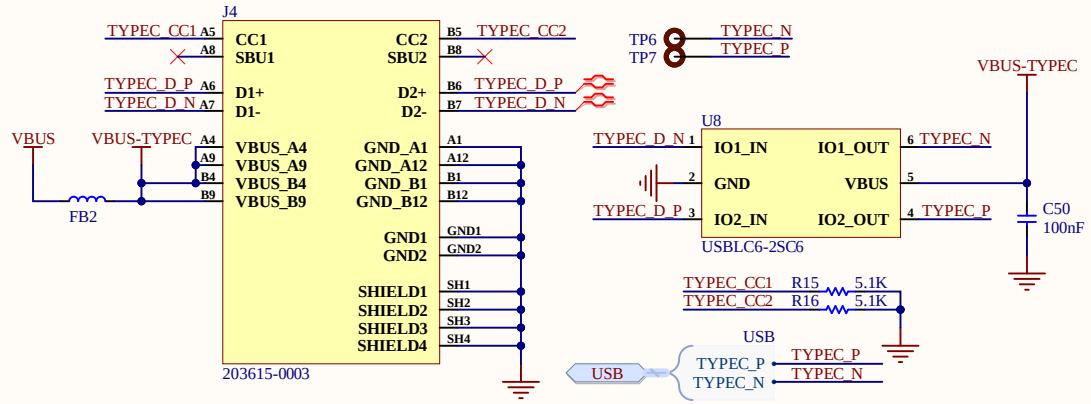
ORING CIRCUIT



RF frontend power switch



USB AND ESD PROTECTION



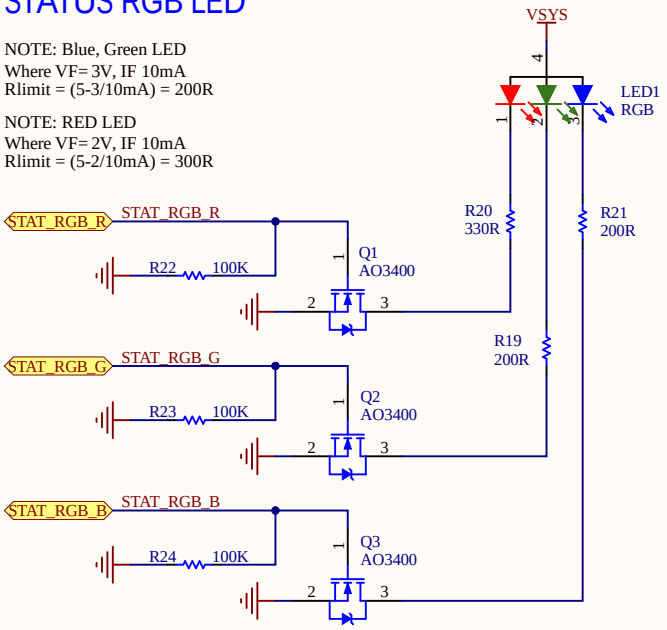
ZyntelliTech

Title: 1.Power Source.SchDoc	
File: Gear_Trak-v1.0.PrjPcb	
Date: 7/16/2025	Page Num: 4
Drawn By: Huzaifa Ali	Page Total: *

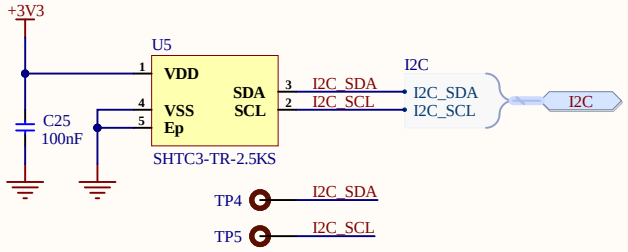
STATUS RGB LED

NOTE: Blue, Green LED
Where VF= 3V, IF 10mA
Rlimit = (5-3/10mA) = 200R

NOTE: RED LED
Where VF= 2V, IF 10mA
Rlimit = (5-2/10mA) = 300R



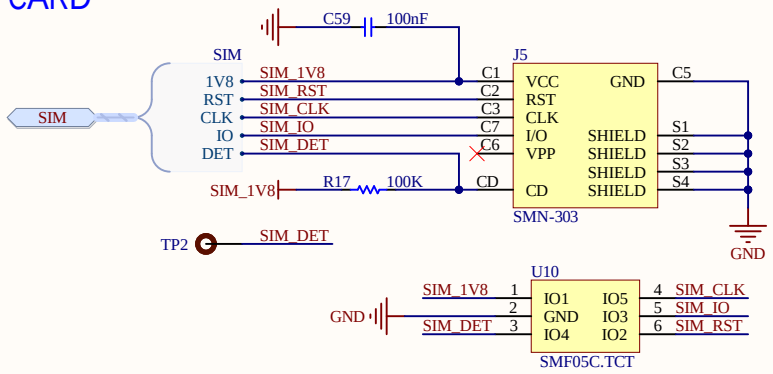
TEMPERATURE AND HUMIDITY SENSOR



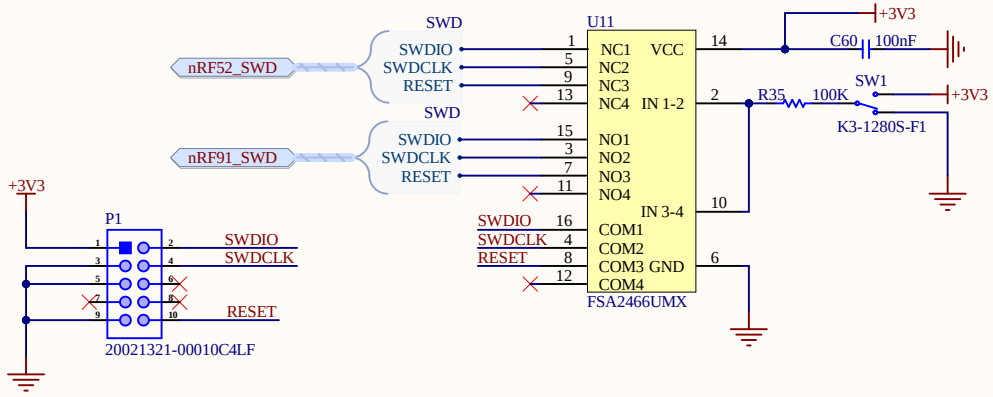
ZyntelliTech

Title: 5.RGB_TEMP-HUMIDITY.SchDoc	
File: Gear_Trak-v1.0.PrjPcb	
Date: 7/16/2025	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: *

SIM CARD



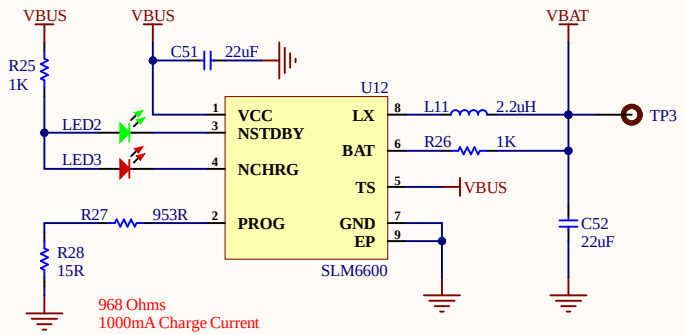
SWD INTERFACE



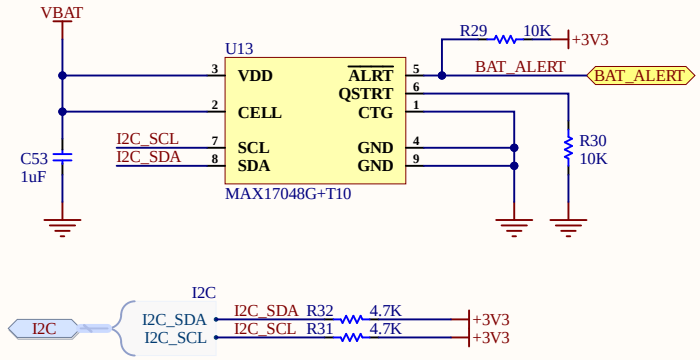
ZyntelliTech

Title: 6.Connectors.SchDoc	
File: Gear_Trak-v1.0.PrjPcb	
Date: 7/16/2025	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: *

BATTERY CHARGER

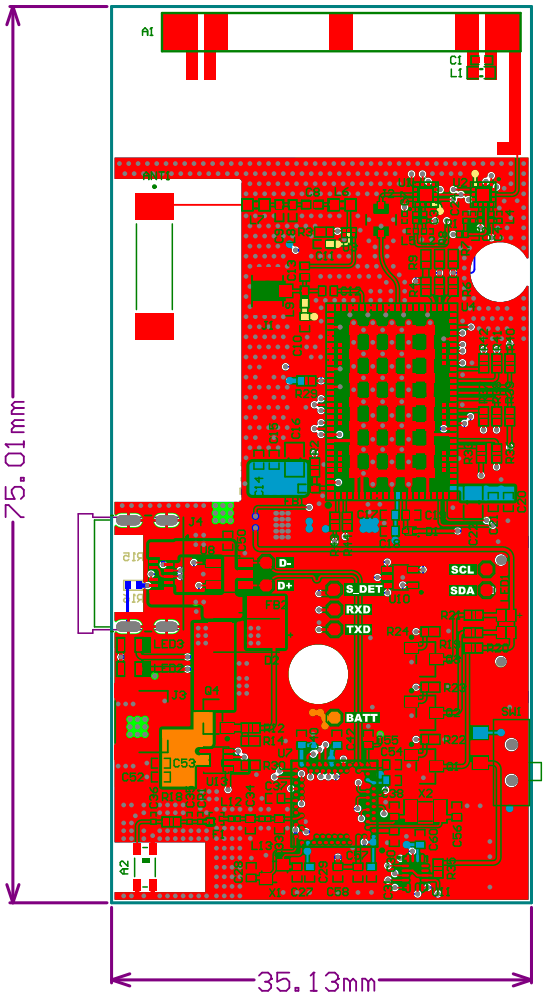


BATTERY FUEL GAUGE



ZyntelliTech

Title: 2.Battery Management.SchDoc	
File: Gear_Trak-v1.0.PrjPcb	
Date: 7/16/2025	Page Num: *
Drawn By: Huzaifa Ali	Page Total: *



Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.030mm	3.5	
1	TOP LAYER		0.035mm		
	Dielectric 1	PP-015	0.099mm	4.1	
2	GROUND1		0.015mm		
	CORE 1	Core-025	0.550mm	4.6	
3	SIGNAL	CF-004	0.015mm		
	Dielectric 2	PP-016	0.116mm	4.16	
4	GROUND2	CF-004	0.015mm		
	CORE 2	Core-025	0.550mm	4.6	
5	POWER		0.015mm		
	Dielectric 3	PP-015	0.099mm	4.1	
6	BOTTOM LAYER		0.035mm		
	Bottom Solder	Solder Resist	0.030mm	3.5	
	Bottom Overlay				

Total board thickness: 1.606mm